2D CAM FOR CNC · IN THE BROWSER

# COPO

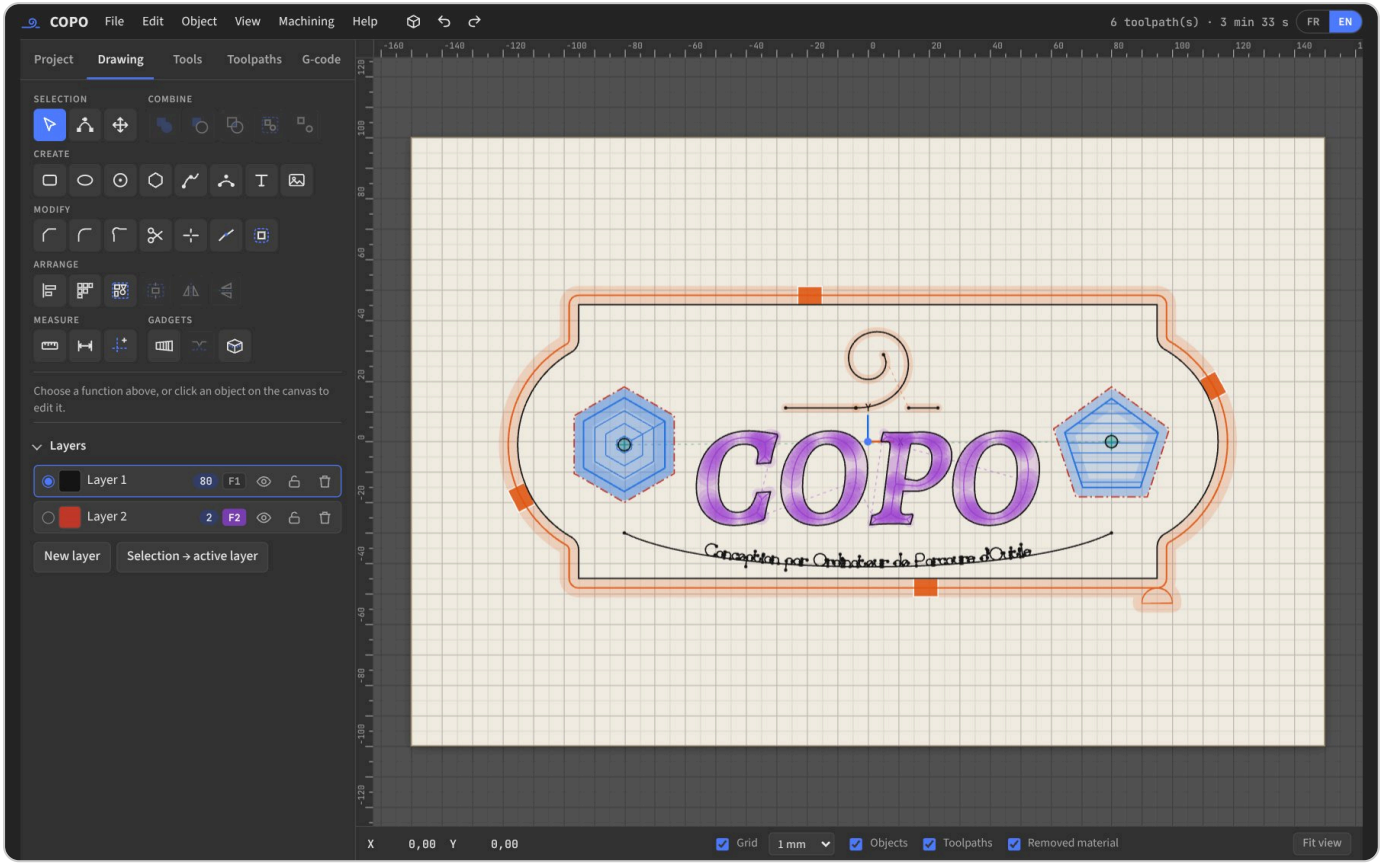
COPO draws your parts as vectors, calculates toolpaths (profile, pocket, V-carving, drilling) and writes the G-code for your machine. Everything runs in the browser, with no installation and no account.

`Version 1.102` `GPL v3` `Windows · macOS · Linux`

`Chrome · Edge · Firefox` `Français · English`

**Open the app**
`https://gbulleryahen-hash.github.io/copo/`

**GitHub repository (code, download)**
`https://github.com/gbulleryahen-hash/copo`




The sample project: COPO text V-carved, a line of single-line text on a curve, a decorative flourish with a ball nose bit, two drilled holes, a cut-out with tabs and two pockets on side 2. Toolpaths are shown in color on the stock.

# At a glance

### Vector drawing
Rectangle, ellipse, polygon, star, arc, Bézier pen, text, nodes, chamfers, fillets, booleans, layers, dimensions.

### 2D machining
Outside, inside or on-the-line profile, ring or raster pocket, rest machining, drilling.

### V-carving
Along the medial axis, like Aspire: the bit rises up to the tip of every corner.

### Tool entries
Back-and-forth ramp, continuous ramp, arc, ramped arc, helix. Allowance and finishing pass.

### Tabs
Spaced automatically or placed by hand, flat or triangular.

### Nesting
Places several parts on the sheet following their true shape, with rotations.

### Single-line fonts
About sixty included (Hershey, EMS, Relief…) for fast one-pass engraving.

### Double-sided
Layers per side, flip around X or Y, pause to turn the part over.

### 3D preview
Material removal simulation, side 1 and side 2, with animated playback.

### G-code
GRBL, grblHAL, manual GRBL (G53), Carbide Motion, Onefinity, LinuxCNC, Mach3/4, UCCNC, MASSO or a custom post-processor. G2/G3 arcs.

### Setup sheet
Printable workshop sheet: 2D and 3D views, origin, tools, toolpaths, speeds, checks.

### Millimetres or inches
Drawing, tools and feeds in mm or inches (fractions available), G-code in G21 or G20.

### French and English
Interface and user guide in both languages, chosen with the FR / EN buttons at the top right.

### Free software
GNU GPL v3 licence: use, modify and share COPO freely.

# Your first job in six steps

1

**Set up the project**

**Project** tab: stock size and material, X0 Y0 origin point, Z zero (top of stock or table), units and machine.

2

**Draw or import**

Draw with the icons of the Drawing tab, import an SVG, a DXF or an image (the **Import** icon in the Create group, or the File menu) or type some text.

3

**Check your tools**

Choose the material in the Project tab, then check in the **Tool library** (Toolpaths tab) the speed, feed, pass depth and stepover of each bit.

4

**Create the toolpaths**

Select shapes on the canvas, open the **Toolpaths** tab, choose Profile, Pocket, V-carving or Drilling, adjust, then click **Calculate**.

5

**Check in 3D**

The **Machining simulation** button (Toolpaths tab) or **View** › **3D machining preview** shows the machined stock. Play the job back to spot a wrong depth or tool.

6

**Save the G-code**

**G-code** tab: choose the post-processor, then **Save G-code**. Always do a dry run before cutting.

# The interface

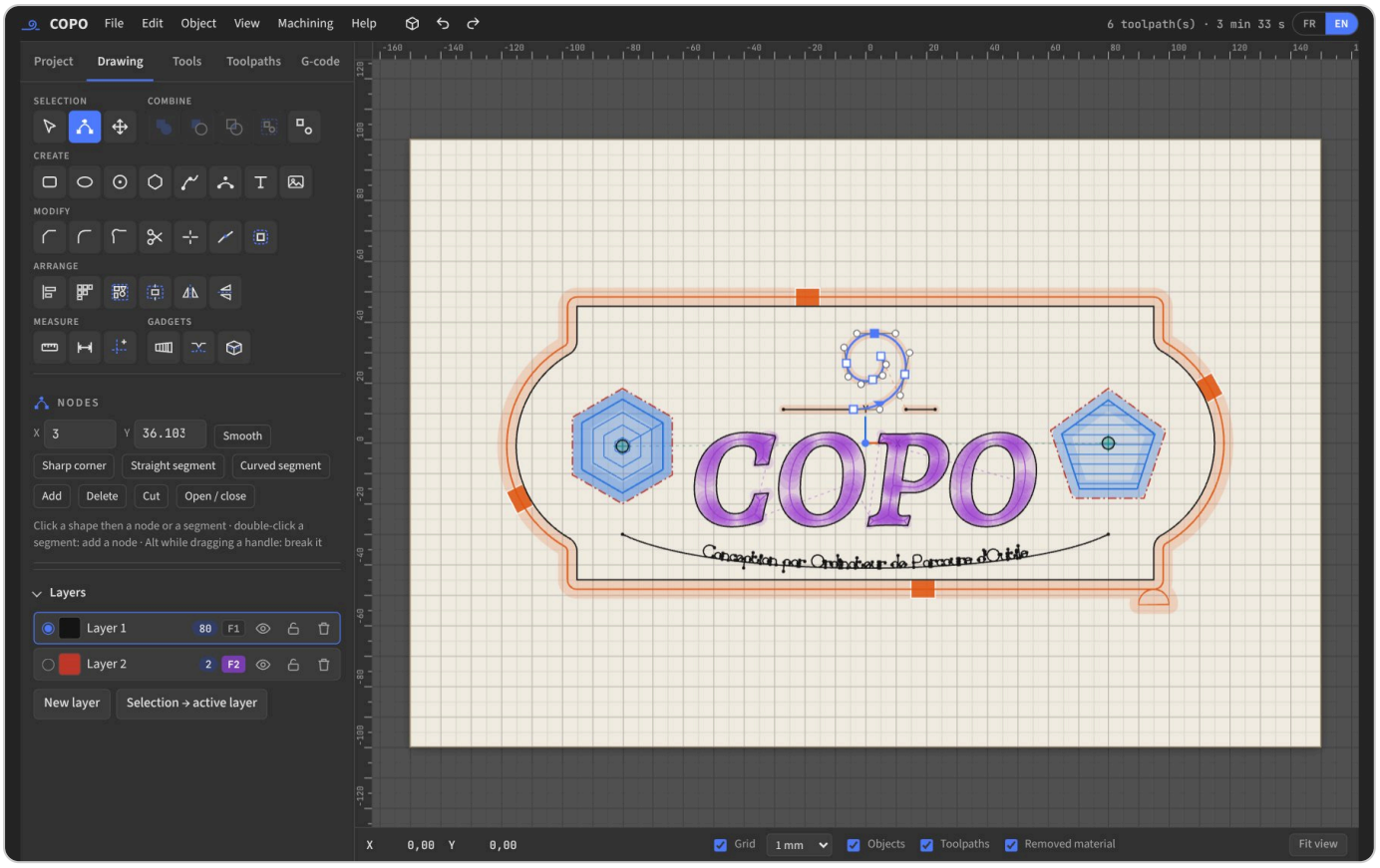
The window reads from left to right, in the order you work.

- **Menu bar**: File, Edit, View, Drawing, Machining and Help. The Drawing menu sorts its commands into six framed categories: Groups, Combine, Vectors, Corners, Arrange and Gadgets. The Help menu opens this guide, lets you report a bug and shows the licence. The **FR / EN** buttons, at the far right of the bar, switch the interface language; the button next to them puts COPO in **full screen** (also View › Full screen; Esc to leave).
- **Drawing tab**: at the top, a grid of icons gathers every drawing function (Selection, Combine, Create, Modify, Arrange, Measure, Gadgets), each with a tooltip and a shortcut letter. Just below appears what you need: the settings of the chosen tool (with a hint and a small diagram), the form of the clicked function (Align, Array copy, Nesting, Images…), or the Selection block when an object is selected. Greyed icons need a selection first. Layers stay at the bottom. Esc closes the open function.
- **Workspace**: the stock in beige, rulers in millimetres, the grid and the origin (X and Y arrows).
- **Left panel** with five tabs: **Project**, **Drawing**, **Relief**, **Toolpaths** and **G-code**. They are numbered 1 to 5 in the usual order of work. In the Project and Toolpaths tabs and in the tool library, small diagrams show what each setting does (depth, allowance, cut direction, tool entry, tabs, bit shape, Z zero…) and update with your choices.
- **Workspace bar**, floating at the bottom left of the 2D view: cursor coordinates relative to the origin, then two groups of on/off icons (their name shows on hover). *Snap* (magnet): grid and its spacing, objects, guides, and the settings button that opens Snap settings just above it. *Show* (eye): grid, toolpaths, removed material. An active button has a blue outline. At the bottom right, the **Recentre** button fits the view to the stock and drawing; the 3D view has the same bars in the same places.
- **Tooltips**: rest the mouse a quarter of a second on a button or setting to see what it does. The tooltip disappears as soon as the mouse leaves, and shows straight away on the next button while you explore. In the toolpath list only the name is shown: the operation, tool, time and any warnings (*!* badge) appear on hover.

A vector is selected by clicking **on its line** (not inside it), so stacked shapes stay reachable. An already selected shape can be moved by dragging from inside.

A floating bar appears under the selection: **Duplicate**, **Group**, **Weld**, **Machine…** and **Delete**. The project is saved automatically in the browser; **File** › **Save project** writes a copy to a `.copo` file (named after the project). Older `.json` files still open; you can also drop a `.copo` file on the workspace to open it.

# Drawing




The Nodes tool on the decorative flourish: the options bar turns a node into a sharp or smooth corner, a segment into a line or a curve, and adds, deletes or cuts a node.

## Shapes and paths

- **Rectangle** `R`, with sharp, rounded or chamfered corners. **Ellipse** `E` (Shift: circle) and **Circle from center** `C`.
- **Polygon and star** `O`: number of sides or points, inner radius.
- **Pen** `P` (first tool of the Create row): click for a corner point, click-drag for a curve point, click the first point to close, double-click to finish. The `space bar` ends the current line and starts a new one; a *right click* ends the line and leaves the pen. The **Next point** block in the panel creates points from the keyboard: type *X* and *Y* (absolute position), or an *angle* and a length *L*, or *dx* and *dy* from the last point. The six fields stay linked (typing the angle and length fills in X, Y, dx and dy) and a dashed preview shows the point. `Enter` or **Add** places it, **Undo** removes the last point, **Close** and **Finish** end the path.

- **3-point arc** `A`: start, end, then the bulge.
- **Nodes** `N`: move nodes and handles, double-click a segment to add a node, `Alt` while dragging a handle to break it. Three node types: **Smooth** (circle, aligned handles), **Cusp** (diamond, two independent handles for a corner between two curves) and **Sharp corner** (square, no handles). Several nodes: `Shift` +click to add or remove one, drag a box in empty space to take them all at once ( `Shift` to add), `Ctrl+A` for all. Selected nodes move together with the mouse or the **arrow keys** (1 mm, `Shift` 10 mm, `Alt` 0.1 mm); Smooth, Sharp corner and Delete apply to all of them. A selected **segment** (click it, it turns red) is deleted with **Delete** or the `Del` key: a closed outline becomes an open line, an open line is split in two. To switch to another shape, click its outline. **Explode** (also in the Modify row) cuts the profile at every node: each segment becomes a separate vector.

## Editing

- **Chamfer** `K` and **Fillet** `F` on one corner, or on every sharp corner from the **Corners** block.
- **Trim** `X` deletes the part of a vector between two intersections; **Cut** `U` splits a vector where you click.
- **Weld, Subtract, Intersect** for boolean operations; **Join** and **Close** for open vectors, gaps being bridged by a straight line or by a smooth curve that continues both vectors; **Offset contour**: on a closed vector outside, inside or both sides; on an open vector to the right or left of the path, or all around (“Both” creates a closed outline). Sharp or rounded corners, and an option to delete the original.
- **Smooth (fit curves)** (Modify): replaces vectors made of many small segments (tracing, DXF, import) with *arcs*, *Bézier curves* or *straight lines* that stay within the tolerance of the original. “Keep sharp corners” keeps as sharp corners those whose change of direction exceeds the max angle (60° by default). A dashed orange preview and the node count before/after are shown before applying; untick “Replace” to keep the originals. *Arcs* or *Bézier*? In COPO every curve (circles, arcs, ellipses, letters) is stored as Bézier curves, as in an SVG file. The *Arcs* mode finds the portions that follow a true circle (constant centre and radius) and draws them with Béziers fitted to that arc, at most one per quarter circle: the gap from a perfect arc is about 0.03% of the radius (under a hundredth of a millimetre for a 30 mm radius). Use it for mechanical shapes (templates, fillets, holes): the radii are truly constant and give clean G2/G3 arcs. The *Bézier curves* mode gives free-form curves, often with fewer nodes on organic shapes (traced drawings, scrolls), but a “round” part is not necessarily an exact circle.

- **Align and distribute**: choose the reference in “Align to”. *The selection*: a single object aligns to the stock, several align to each other. *The first selected object*: it stays put and the others align to it; Distribute places them after it, in the order you selected them, with the given spacing. *The stock*: aligns to the stock edges and distributes across its full width or height.
- **Rotation** (Arrange): an orange pivot appears at the centre of the selection; drag it on the plan (it snaps to objects, guides and the grid), pick one of the nine points, or type its coordinates. *Relative* rotation (the selection turns by the given angle, positive counter-clockwise) or *absolute* (the selection takes the given angle; COPO remembers earlier rotations). ↺ 90° and ↻ 90° buttons for quarter turns.
- **Handles**: click an already selected object a second time. Squares resize (corners keep proportions when “Keep proportions” is ticked, Shift for the opposite; middles resize one axis), circles rotate around the pivot (Shift: 15° steps), and the pivot can be moved. Click again to hide them.
- **Right-click and double-click**: a single right-click on the plan finishes the current line (polyline), cancels a shape in progress, goes back to the selection tool and deselects; with the selection tool, it hides the handles and then deselects. Double-clicking an object, from the selection tool or a shape tool (rectangle, ellipse, circle, polygon, arc, dimension, measure), selects it with its handles; another double-click opens node editing (Esc or right-click to leave).
- **Array copy** in a grid, a circle or *along a curve* (also the **Repeat along a curve** icon in Arrange): select the objects to repeat then, last, the guide curve (`Shift`+click); set the number of copies, spread over the whole curve or with a spacing, a start offset, and tick *Align to the curve* so they rotate with it. **Mirrors**, **rotation**, **stacking order**.
- **Exact position and size**: **Selection** block, with a reference point chosen among nine (corners, middles, center).

## Snapping, guides and dimensions

**Edit** › **Snapping…** opens the **Snap settings** window: snap to grid and its spacing, **guides**, stock corners, midpoints and centre, snap radius (small to large), and object snapping with its five points (ends and nodes, midpoints, intersections, centres, tangents). These settings are remembered. Drag from a ruler to create a guide: drawn points snap to it, and when you move a selection its edges and centre line up with the guides (and with the stock if that box is ticked). An orange symbol at the cursor shows the snap found. **View** › **Show grid** shows or hides the canvas grid. The **Measure** tool `M` gives a distance and the **Dimension** tool `D` places a dimension that stays on the canvas. An end snapped to a node, a segment midpoint or the centre of an arc or circle (solid dot at the end of the extension line) stays attached to the object: move it, resize it or edit its nodes and

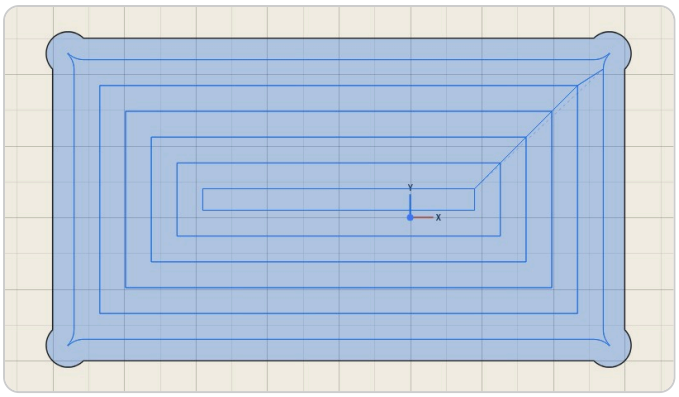
the dimension follows, with its value updated. An end placed elsewhere (grid, intersection, free point) stays fixed.

## Layers, images and files

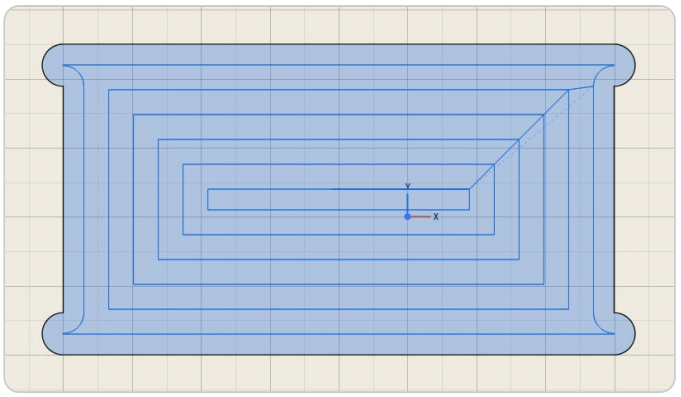
- **Layers**: stroke color, visibility, lock and machined side (F1 top, F2 bottom).
- **Reference images** (**Import** icon › Image, **File** › **Insert an image…** or drag and drop): move, resize and rotate them with the handles to trace over them. They are never machined.
- **Vectorize an image** (Drawing panel › Images, image selected): COPO traces the outlines of the image. Set the *threshold* (anything darker becomes the shape; “Auto threshold” works it out for you) and the *smoothing* (softer outlines, small details removed), tick *Invert* to outline light areas. The preview updates live; **Create outlines** adds closed, grouped shapes where the image is, ready for a pocket, a profile or V-carving. A sharp, high-contrast image (logo, line drawing, silhouette) gives the best result.
- **Import**: the **Import** icon in the Create group opens a panel with two choices, *Vector* (SVG, DXF) and *Image* (template to trace over or vectorize).
- **SVG**: import in millimetres with curves kept; export the whole drawing or the selection.
- **DXF** (text/ASCII, any version): lines, polylines with arcs, circles, arcs, ellipses, splines and blocks. The file's unit is respected (mm, cm, inches…); a file without a unit is read as millimetres. Touching segments are joined into outlines, closed when possible, and each DXF layer becomes a COPO layer. Text, dimensions, hatches and 3D objects are skipped (the message says which).

## Corners for tenons and mortises: dogbone and T-bone

A round bit leaves a fillet of its radius in corners, so a square tenon will not fit its mortise. The relief is drawn on the shape, corner by corner, so it only goes where it is needed. Take the **Relief** tool `B` (Modify group), choose *Dogbone* (the bit moves diagonally to touch the sharp corner) or *T-bone* (the relief extends the longer side, less visible) and the **bit diameter**, then click near the corners to relieve. The same settings are in the **Corners** block of the Drawing panel. Corners are picked one by one, as for chamfers and fillets. Relief only works between two straight segments; its diameter is 0.1 mm larger than the bit so it fits without forcing. Then machine the shape as usual (pocket, inside or outside profile) with that bit.

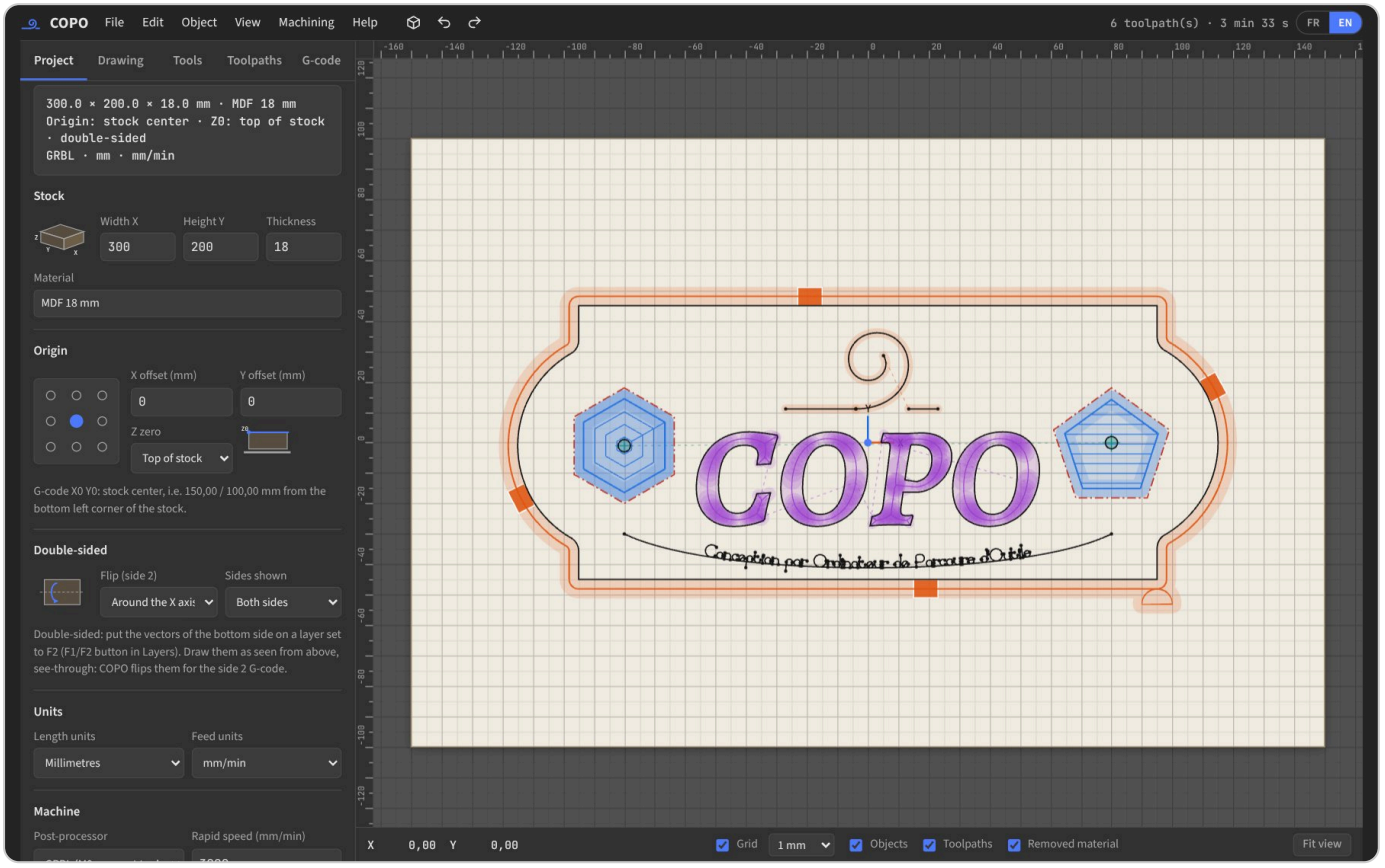


**Dogbone**: an arc the size of the bit, diagonally in each corner.



**T-bone**: the arc extends the longer side.

# Project tab




The Project tab gathers everything you set at the start of a job. The summary at the top lets you check it all at a glance.

The **Project** tab, first in the left panel, opens by itself when you create a new project. It contains:

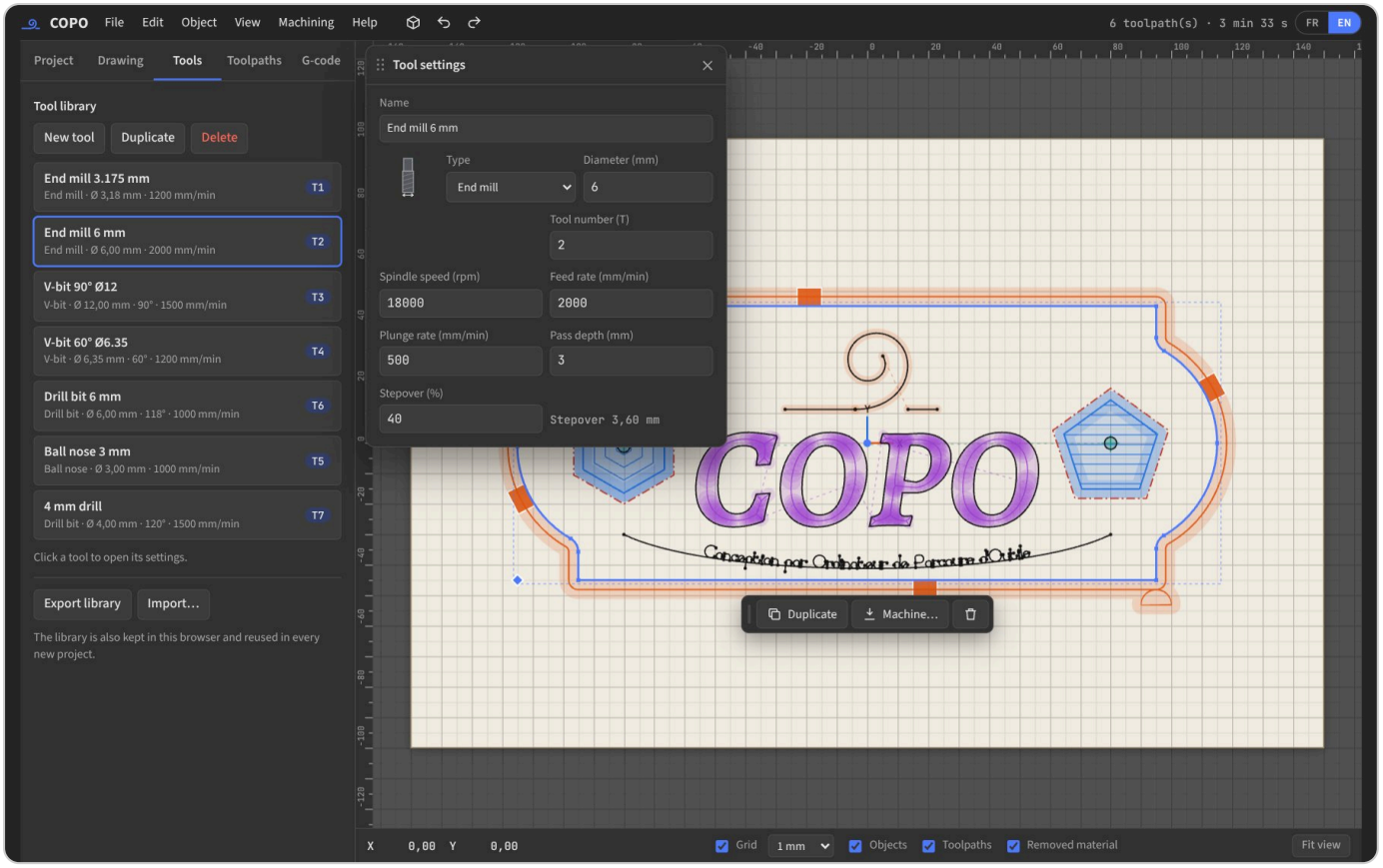
- **Stock**: X width, Y height, thickness and material (used by the setup sheet);

- **Origin**: the X0 Y0 point on the nine-point grid, an X/Y offset to place it outside the stock, and Z zero on the top of the stock or on the table;
- **Double-sided**: the flip axis and the sides shown;
- **Units**: millimetres or inches, feed unit;
- **Machine**: post-processor, rapid speed, **maximum cutting feed** (2000 mm/min by default: tool feeds are capped at this value), safe and clearance Z, G2/G3 arcs and comments. The **G81/G82/G83 drilling cycles** box writes holes as canned cycles: G81 plain, G82 with a dwell at the bottom, G83 with pecking, G98 retract to clearance Z. G83 has no dwell: a hole with both pecking *and* dwell is written as G0/G1. It is ticked by default for grblHAL, LinuxCNC, Mach3/4, UCCNC and MASSO, and unticked for GRBL, Carbide Motion, Onefinity and the custom post-processor (holes are then written as G0/G1); you can always change it. If the controller does not know these cycles, it simply stops with an error.

The origin point must match the point where you set X0 Y0 on the machine. Stock, origin and material are saved with the project; units and machine settings are remembered in the browser.

> Example: with the origin at the bottom left corner, a square whose corner sits at X0 Y0 on the canvas is cut from the bottom left corner of your sheet. The line under the grid tells you where the G-code X0 Y0 falls.

# Tool library




The tool library: the material at the top, tools sorted by unit system and type on the left, the tool's settings for that material on the right.

The library opens from the **Tool library** card in the Tools block of the Toolpaths tab (it shows the project material), from **Machining** › **Tool library…**, and from the **Tool** button of a toolpath window. Inspired by Aspire's, it comes filled: metric (mm) and imperial (inch) tools, sorted into end mills, ball nose, V-bits, engraving bits and drills, with settings for seven materials: **softwood, hardwood, MDF, acrylic, PVC, aluminium and brass**. The first four come from an Aspire library tuned for a WorkBee-type machine; settings missing there (large bits), PVC (like acrylic), aluminium and brass are **estimated** for a hobby CNC, at the low end of the ranges recommended for light machines (Shapeoko, X-Carve, 3018, Stepcraft; cross-checked with Carbide 3D and Sienci): in aluminium about 0.015 mm per tooth at Ø 3 and 0.025 mm at Ø 6, brass 20% lower. Estimated values are flagged by an orange banner: do a test cut and correct them. These values are deliberately cautious; makers' charts, written for rigid machines, go 3 to 5 times further.

- **Material** (top): this is the project material, also set in the Project tab (*Tool material*). Changing it makes every project tool take that material's settings and recomputes the toolpaths. The **+** button adds a material.

- **Tree** (left): click a folder to open it, double-click to rename it; the search box filters by name; “used” marks the tools of the project's toolpaths. At the bottom: new tool, new folder, duplicate, delete and back to the default library (two clicks). The **Import…** and **Export…** buttons are at the top right.
- **Settings** (right): geometry (name, type, diameter, angle, engraving tip, number of flutes, T number, notes), then cutting settings for the chosen material. The name updates itself as long as you have not edited it.

| SETTING | PURPOSE |
|---|---|
| Type | End mill, ball nose, V-bit (V angle), tapered flat-tip engraving bit (angle and tip diameter), **tapered ball-nose** or drill (point angle, usually 118°). |
| Tapered ball-nose bit | Defined as on its box: **shank diameter D**, **tip radius R** and **cutting length L**. COPO derives the side angle, that of the cone tangent to the ball that reaches diameter D at height L: for D 6, R 1, L 20, about 5.99° (11.99° included). Engraving and tapered bits are used like V-bits of that angle (V-carving, on-the-line profile); the rounded or flat tip is not yet taken into account in the depth. |
| Diameter | Used to calculate offsets and pockets. |
| Tool number (T) | Written in the G-code of tool-change post-processors ( `T2 M6` , `M6 T2` ). |
| Spindle speed | In rpm, written with `M3` . |
| Feed, plunge rate | In mm/min, for cutting and for plunging or ramping. |
| Pass depth | Maximum depth per pass; it fills the toolpath's pass depth when you choose the tool. |
| Stepover | Distance between two pocket passes, in mm or % of the diameter (as in Aspire). |
| Number of flutes | Number of cutting edges of the bit. |
| Feed per tooth | Calculated: feed ÷ (spindle speed × number of flutes). Type a value to recalculate the feed. The comment below, on an orange background when conditions are not optimal, says whether it is low (the bit rubs and heats up) or high (risk of breakage), against the hobby-CNC ranges (wood and MDF, plastics, aluminium, brass) by diameter, and warns when the feed exceeds the machine's maximum feed; the bit maker's data come first. For a drill, COPO gives the feed per revolution. |

## Choosing a toolpath's tool

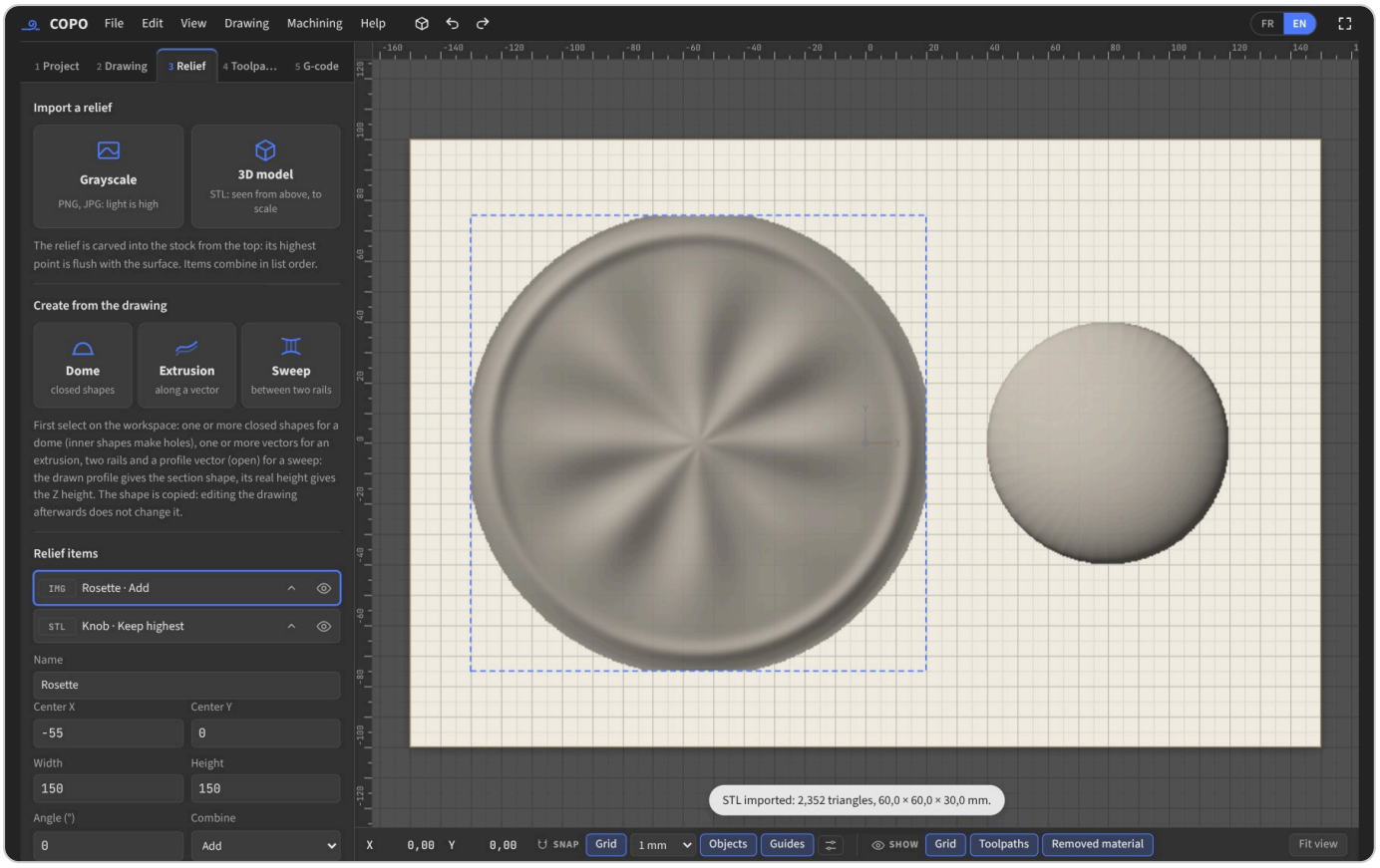
In a toolpath window, the **Tool** button shows the chosen tool; click it to open the library, select the tool then **Use this tool** (or double-click). Tools that do not suit the toolpath type are greyed out. The toolpath takes the tool's settings for the project material, and its pass depth. Editing a tool in the library updates the toolpaths that use it.

## Importing an Aspire library

The import button reads Aspire, VCarve and Cut2D libraries directly ( `.vtdb` file, for example `C:\ProgramData\Vectric\Aspire\V12.0\ToolDatabase\tools.vtdb` , or an export from Aspire's library). Tools, their folders and their per-material settings are added in an “Aspire import” folder, inches converted to millimetres; if the file holds several machines, COPO asks which one to use. Missing materials (PVC, aluminium, brass) are estimated. Form tools and diamond drag bits, which COPO cannot machine yet, are skipped. Import also accepts a COPO library exported as `.json` .

# Relief (modelling)

The **Relief** tab prepares sculpted machining, like Aspire's modelling: each relief item is a height map placed in the stock. The tab opens the 3D view next to the drawing (**2D** / **3D** buttons); the creation buttons show how to use them on hover.




A grayscale image (rosette) and an STL model (knob, "keep highest") with the item settings.

- **Grayscale**: insert the image (PNG, JPG) in the Drawing tab (**Import › Image**), place and size it, select it, then click **Grayscale**: the image is *turned into a relief* at the same place, size and angle, and leaves the 2D view. The relief is shown shaded on the workspace and handled like an object. Light is high, dark is low (tick *Invert* for the opposite). A transparent background counts as black for heights but is not machined: with no boundary selected, machining stays within the opaque part of the image.
- **3D model** (binary or text STL): the model is seen from above and projected into heights, at the file's scale (reduced if larger than the stock). *Model top* picks the axis facing up (+Z, −Z, ±Y, ±X). Undercuts are machined as seen from above.
- **Settings of each item**: position and size as for an object (one of nine reference points, X, Y, width, height), *relief height*, *stock top → relief top* and

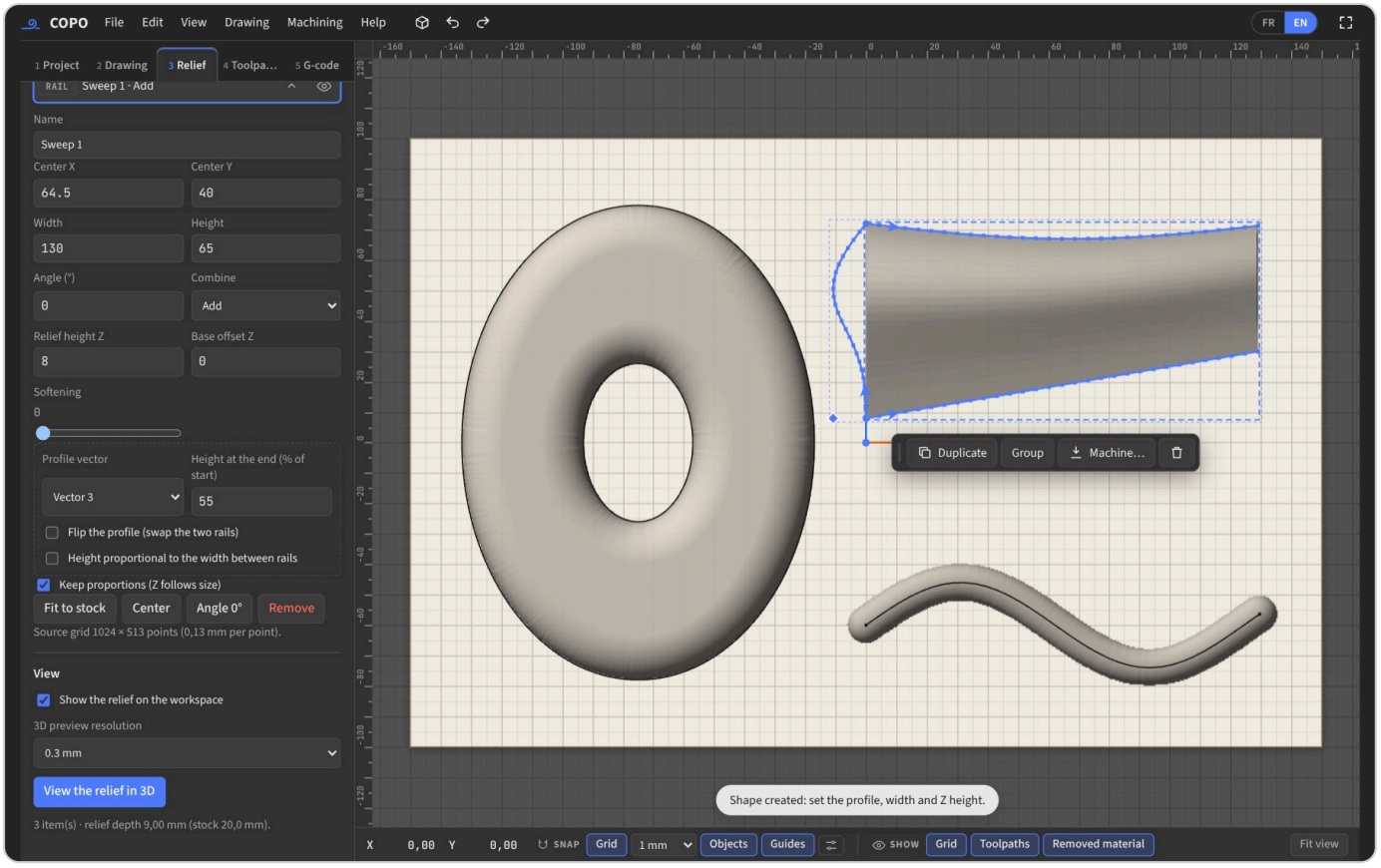
*stock bottom → relief bottom* (editing one recalculates the other: their sum plus the height is the stock thickness), angle, *softening* and how it *combines* with the items before it: add, subtract, keep highest or keep lowest, or *blend*: where reliefs overlap, their heights are averaged, faded towards the item's edges, without adding or subtracting. "Keep proportions" makes the height follow size changes.

- **Limit with vectors**: click **Limit with vectors…**, pick one or more closed vectors on the workspace and click **Apply the limit**. The relief only exists inside those vectors (a shape inside another makes a hole); outside there is no relief. The limit follows the vectors if you edit them; **Remove the limit** gives the whole frame back to the relief.
- **On the workspace**, an STL model or a grayscale relief is handled like an object with the Select tool: a first click selects it and opens its position settings in the Relief tab, and you can drag it to move it; a second click shows the handles: squares to resize (corners: proportions kept, Shift to distort; sides: one dimension only), circles to rotate (Shift: 15° steps). Delete removes it. Domes, extrusions and sweeps through the 3D view or the panel.
- **Display**: in 2D, STL models and grayscale reliefs are shown shaded (domes, extrusions and sweeps keep their vectors); the whole relief is visible in the 3D view. The summary gives the depth of the lowest point and warns if it exceeds the stock thickness.

Relief data is kept in the browser and saved in the `.copo` file.

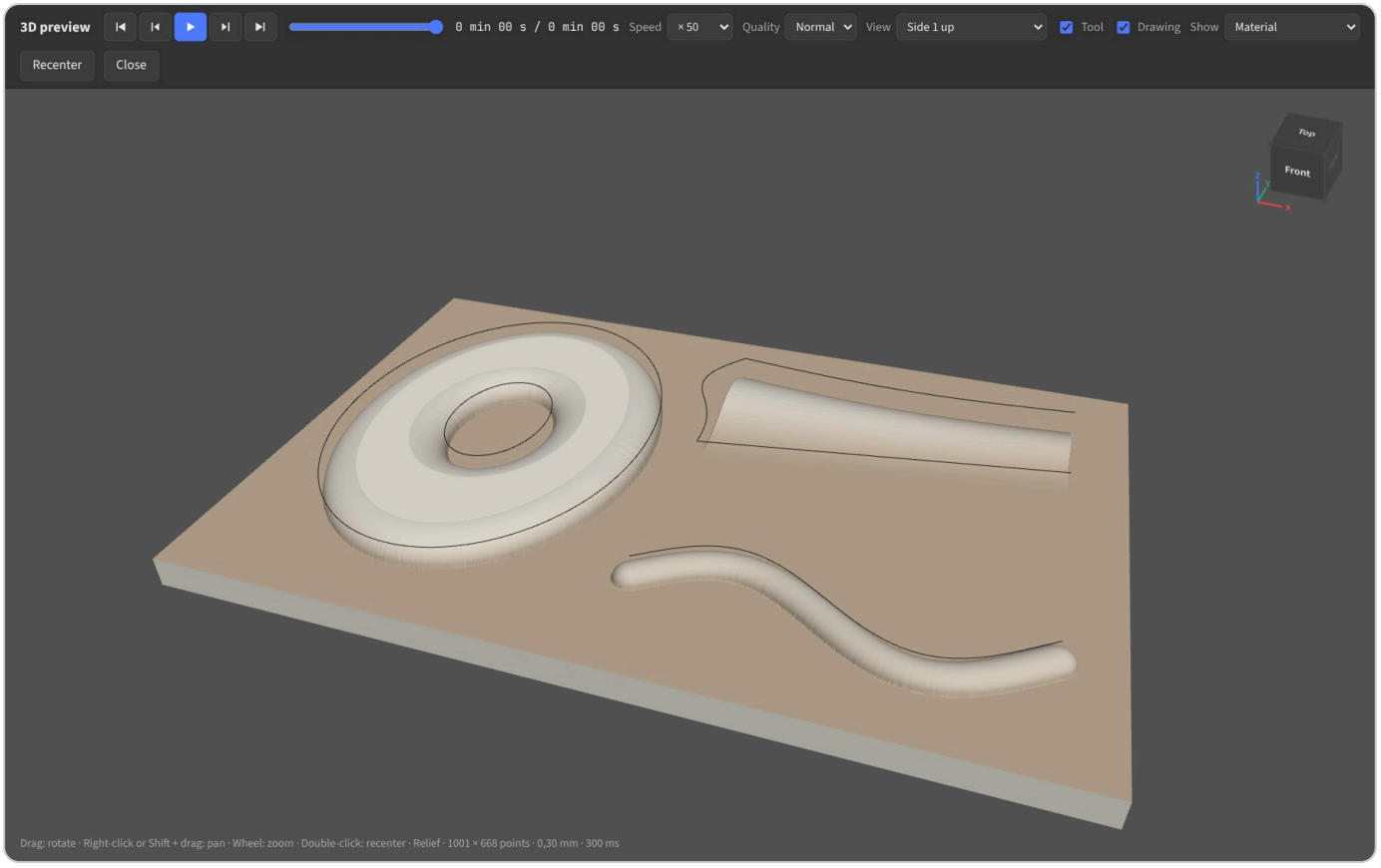
## Shapes built from the drawing

Select vectors on the workspace, then click one of the three buttons under **Create from the drawing**. The shape is copied into the relief: editing the drawing afterwards does not change it.




A dome with a 14 mm edge on an ellipse with a hole, an extrusion along a curve and a two-rail sweep with an ogee profile (end height 55%).

- **Dome** (one or more closed shapes; inner shapes make holes): height rises from the edge. *Edge width* = 0 domes up to the center (a ridge along elongated shapes, as in Aspire); a value gives a rounded edge of that width, then a plateau.
- **Extrusion** (one or more vectors, open or closed): a moulding of the chosen *width* follows the path, with rounded ends.
- **Sweep** (two rails + one profile vector, all open): the drawn section spans from one rail to the other, scaled to the gap between them. The profile is the vector joining the two rails, otherwise the last one selected (changeable with *Profile vector*); its chord runs from rail to rail and its real height gives the Z height; *Flip the profile* reverses it if needed. Without a third vector, the section is round; *Height at the end* varies the height from start to end (scroll, neck, domed edge) and *Height proportional to the width* adapts it to the gap between the rails.
- **Profile** of domes and extrusions: round, angular (pointed), flat or ogee (S-shaped). Z height, base offset, combine mode and softening work as for other items.




**View the relief in 3D**: the relief carved into the stock, its highest point at the surface.

## Machining the relief

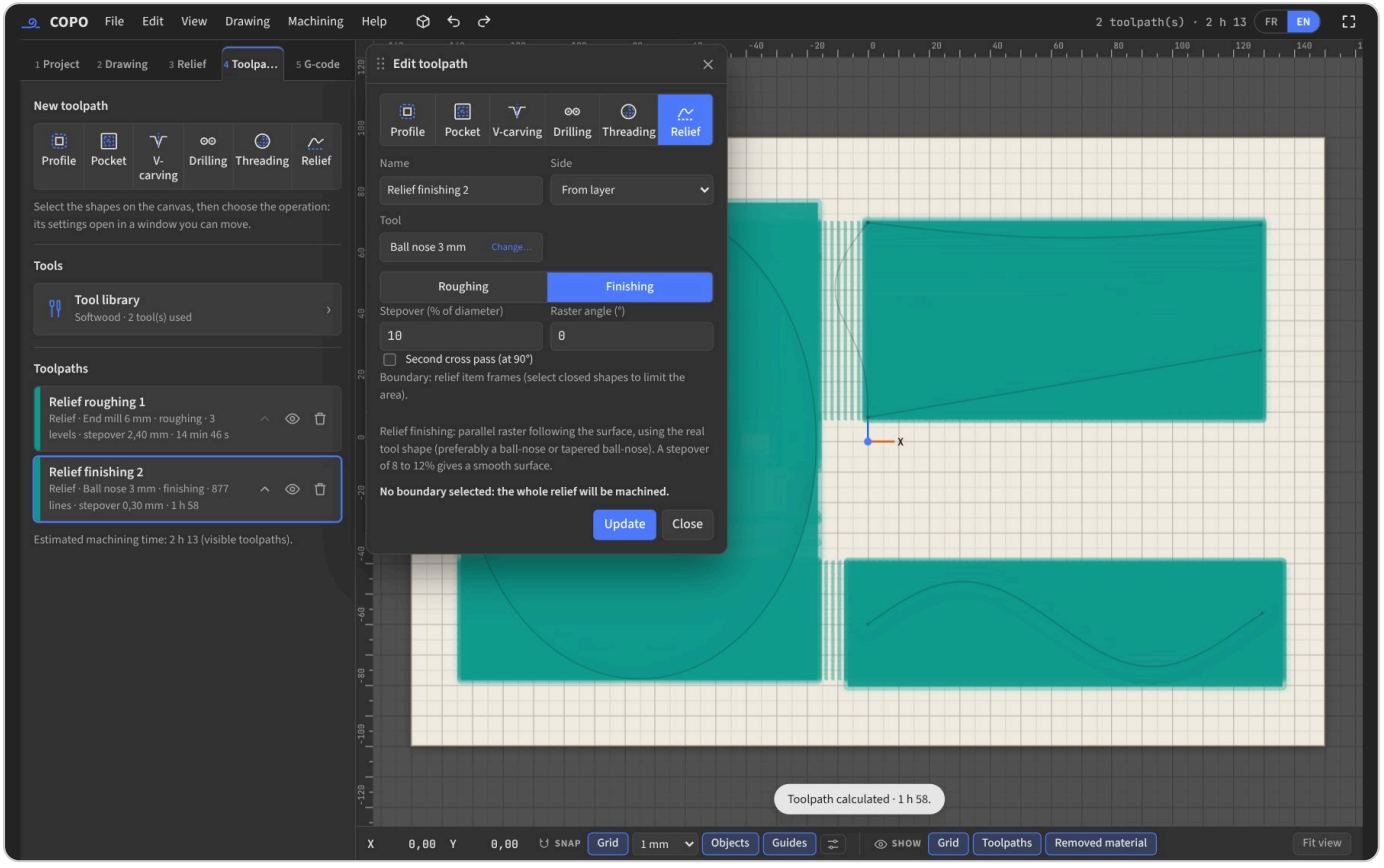
In the **Toolpaths** tab, the **Relief** type offers two operations:

- **Roughing**: raster by Z levels with a flat end mill (the largest is suggested), each level stepping down by the pass depth, leaving an *allowance* (0.5 mm by default) for finishing. Default stepover: 40% of the diameter.
- **Finishing**: parallel raster following the surface with the real tool shape, never gouging the relief. Use a ball-nose, a tapered ball-nose (defined by D, R and L) or, for fine details, a flat-tip engraving bit (angle and tip diameter). A round-tip engraving bit is described as a tapered ball-nose, using the box dimensions: shank diameter D, tip radius R, cutting length L; COPO derives the angle (D 6, R 1, L 20: 5.99°). Default stepover: 10% of the diameter; adjustable raster angle and an optional *second cross pass* at 90°.
- **Reliefs to machine**: tick the items whose outline limits machining (all by default). Heights stay those of the full relief.
- **Machining boundary**: *Model boundary* (the ticked reliefs, without the transparent background of images), *Stock boundary* (the whole stock) or *Selected vectors* (closed shapes selected on the workspace). *Tool relative to boundary*: *tool edge inside* (default, the bit does not cross the vector) or *tool center on boundary* (the bit overhangs by one radius, useful to finish a relief

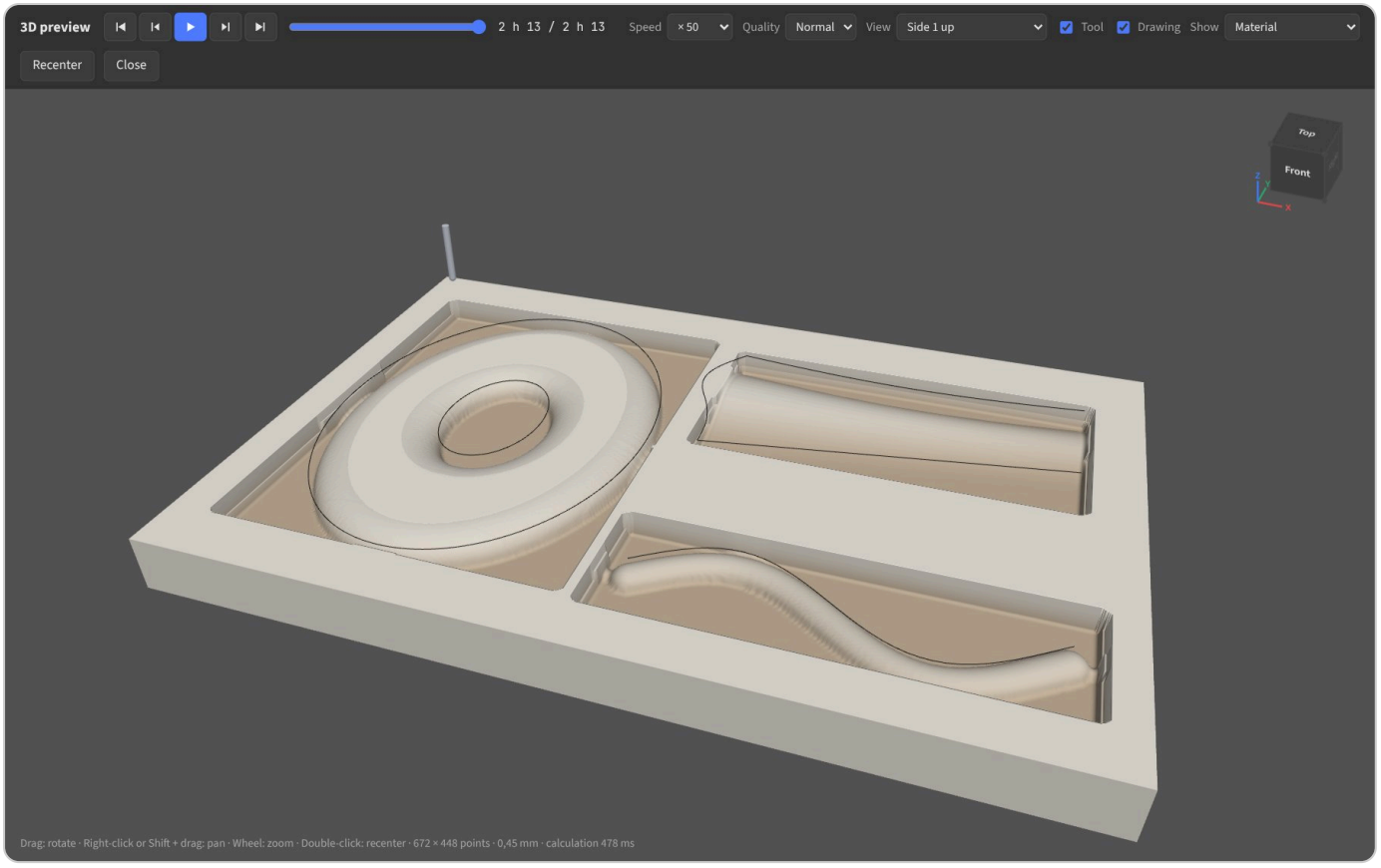
edge cleanly). *Boundary offset* enlarges (positive) or shrinks (negative) the boundary.

- **Strategy**: for roughing, *Z levels* (machines only below each level, in runs) or *3D raster* (each line follows the surface without going below the level, fewer retracts); for finishing, *Raster* (parallel lines at the chosen angle, optional cross pass) or *Offset* (contours parallel to the boundary edge, from outside in, climb or conventional).

Passes are chained in zigzags region by region and short links stay on the surface, to limit retracts and air moves. If the relief changes after calculation, the toolpath says so in orange: open it and click **Update**. Finishing files are long (tens of thousands of lines), as in any carving software; the 3D preview replays roughing then finishing.

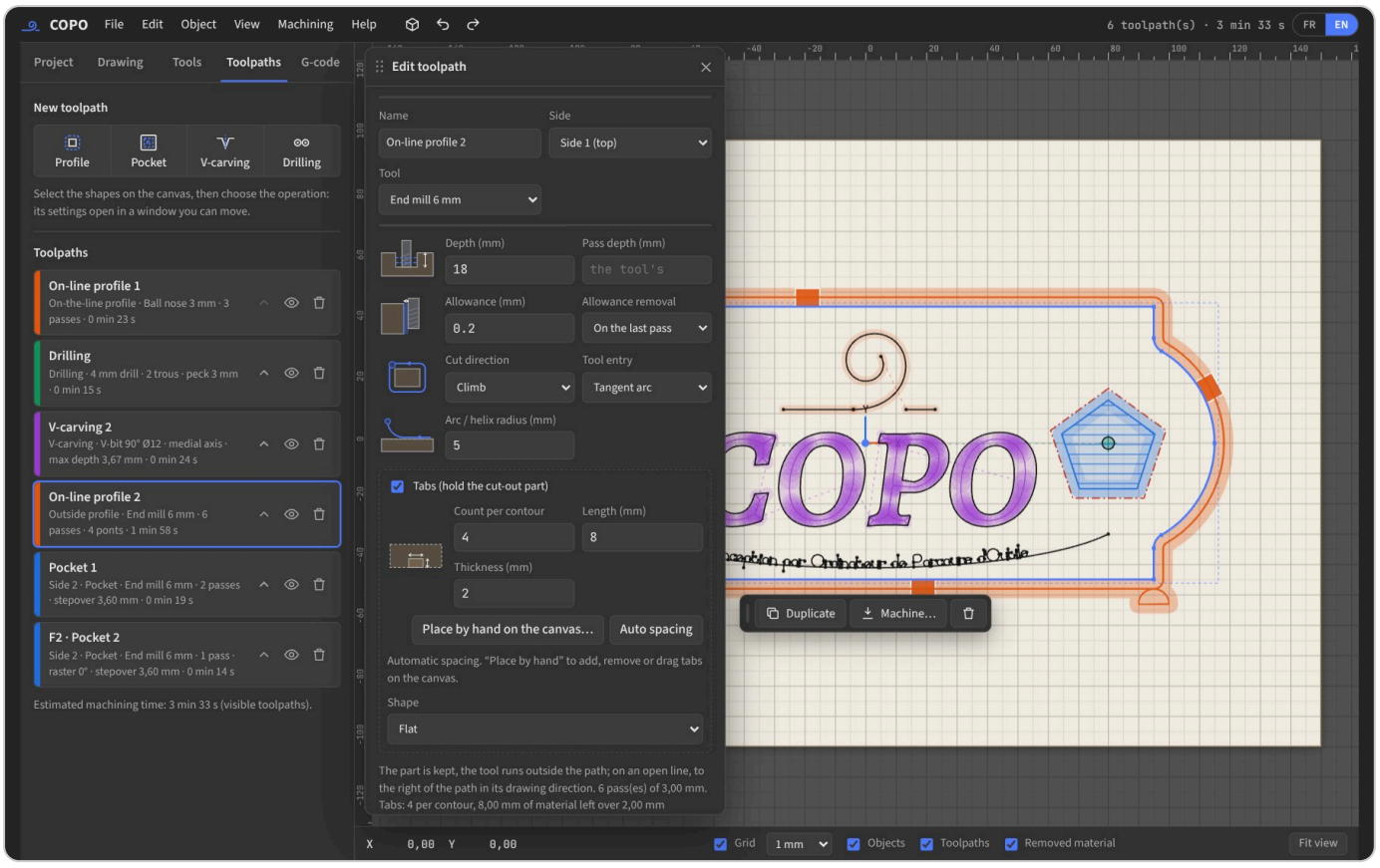



Roughing with a 6 mm flat end mill then finishing with a 3 mm ball-nose, with no boundary selected.




The result in the 3D preview.

# Toolpaths




Click a toolpath in the list to edit it: its path is highlighted on the canvas. The text under the form sums up what the machine will do.

At the top of the tab, the four buttons **Profile**, **Pocket**, **V-carving** and **Drilling** create a toolpath: select the shapes on the canvas (even with the window open), then click the operation. Its settings open in a floating window you can move by its title; COPO remembers its position. Clicking a toolpath in the list opens the same window to edit it.

While the toolpath window is open, the rest of the drawing and the other toolpaths are **dimmed**: only the selected vectors and the current toolpath stand out. Everything returns to normal when the window closes. At the bottom of the window, a **status line** says what will be machined, for example “3 vectors selected (2 closed, 1 open)”; it turns red when nothing is selected, and warns when a pocket, V-carving or drilling will ignore open vectors.

## The four types

- **Profile**: *Outside / right* (the part is kept), *Inside / left* (the hole is cleared) or *On the line*. On an open line, Outside runs the tool to the right of the path and

Inside to the left, in its drawing direction (Drawing › Reverse direction to change it); “On the line” follows the stroke or runs left or right according to the “Open lines” setting. On an open line offset to the right or left, ramp, tangent arc and ramped arc entries apply too: the arc enters and leaves on the waste side.

- **Pocket**: in *rings* parallel to the contour or as a *raster* at the chosen angle. *Rest machining* uses a smaller bit for the corners a larger bit could not reach: keep the same depth and place this toolpath after the original pocket.
- **V-carving**: the V-bit follows the medial axis of the path; it goes deeper where the stroke is wide and rises up to the tip of every corner. A maximum depth gives a flat bottom.
- **Drilling**: one hole at the center of each closed shape, with optional pecking and dwell at the bottom (0 = none, the default). With a flat end mill, circles larger than the bit can be milled helically to the drawn diameter; the option is hidden for a drill bit, a ball-nose or a V-bit. When it is unticked, an orange warning points out circles larger than the bit, which will be drilled at the bit diameter (handy when you draw large circles just to see and select the holes more easily). With a drill bit, each hole is drilled at the bit diameter; COPO shows the tip length to add for a through hole.
- **Threading**: tapped holes and screws milled with a *thread mill* (single-form, 60°), centered on each selected circle. Set internal (tapped hole) or external (screw), pitch, nominal diameter (0 = the circle’s: draw an 8 mm circle for M8), thread hand and number of radial passes; the thread length is the cut depth. COPO shows the pre-drill diameter (nominal Ø − pitch, e.g. Ø6.75 for M8 × 1.25). The bit always climb-mills, one helix turn per pitch, with tangent half-circle lead-in and lead-out; the feed entered is the cutting-edge feed and COPO corrects it at the tool center. For a screw, clear the material around the cylinder first. Two thread mills (Ø4 and Ø8 mm) were added to the library with estimated settings.

## Passes, allowance and cut direction

**Start depth (D)** and **Cut depth (C)**, as in Aspire: machining starts at depth D below the surface (for example at the bottom of a pocket already cut) and goes down by C from there, so the bottom is at D + C. Leave D at 0 to start from the surface. In V-carving, the pattern is computed as if the surface were at D, which is what inlays use. The small diagram next to the fields shows what D and C are.

**Project onto the relief** (profile, pocket, V-carving): the toolpath follows the surface of the Relief tab instead of the flat stock top; depths D and C are then measured from the relief. The contact height takes the cutter shape into account (V, ball-nose, flat), to carve a pattern or lettering on a domed surface. Outside the

relief, the toolpath stays measured from the stock top. Side 1 only; if the relief changes, the list flags that the toolpath must be recalculated.

The cut depth is split into passes of the chosen depth. An **allowance** leaves a little material on the walls while roughing; it is removed on the last pass or by a full-depth finishing pass. The cut direction is *climb* or *conventional*.

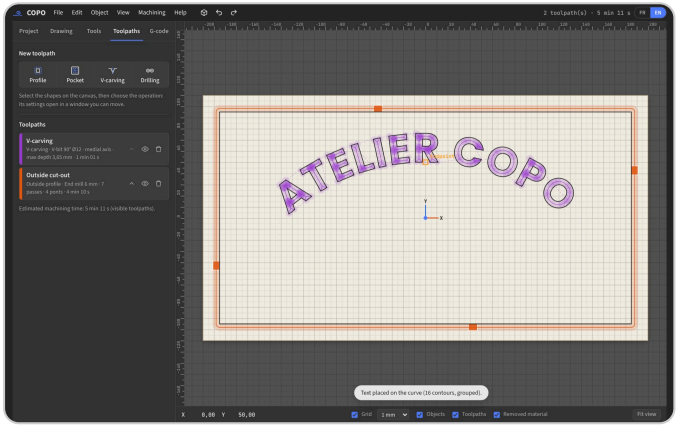
### Tool entry

| ENTRY | BEHAVIOUR |
|---|---|
| Back-and-forth ramp | The tool goes down while moving, returns to the start at full depth, then goes around. Each pass starts and ends at the same point. Default setting. |
| Continuous ramp | The tool goes down while moving and continues around; the ramp portion is recut at the end. |
| Tangent arc, ramped arc | Arc lead-in and lead-out on the contour, to avoid a mark on the finished wall. |
| Helix | The tool spirals down, ideal for pockets and holes. |
| Straight plunge | Vertical plunge, for center-cutting bits or V-carving. |

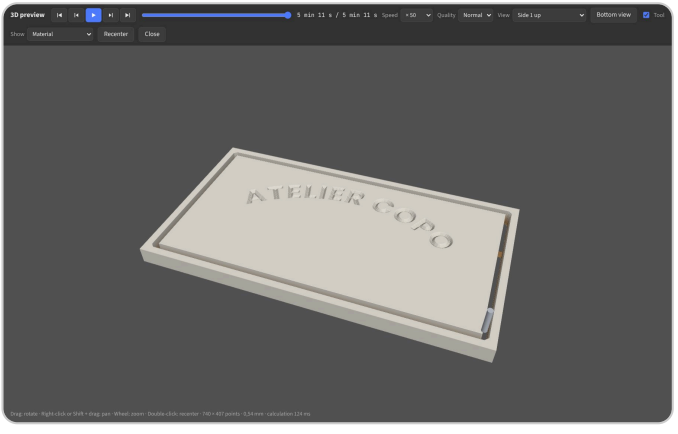
### Tabs

Tabs hold a cut-out part until the end. Set their count per contour, length and thickness, flat or triangular (easier to break). **Place by hand on the canvas…** lets you add a tab with a click on the toolpath, drag it or remove it.

# Text and V-carving




Text placed on an arc, then V-carved.




The same job in the 3D preview.

Choose a font in the **Text** block list: installed fonts (Chrome and Edge) or your fonts folder (**File** › **Open a fonts folder…**), with a preview. Two boxes at the top of the list, *COPO fonts* and *Computer fonts*, shorten the list; the choice is remembered. Take the **Text** tool `T` and click on the canvas. With **On a curve** placement, click a contour: the text centers on it, above or below, with the spacing you want. The text becomes grouped vectors, ready for V-carving, pocketing or profiling. In **Between two curves** placement, click the bottom curve (the baseline) then the top one: the letters stretch to fill the space between them, like an envelope (two arcs for an arched sign, for example).

### Single-line fonts

COPO comes with about sixty **single-line fonts** (Hershey, EMS, Cutlings, Relief SingleLine, Routed Gothic…), stored in the `fonts/trait-unique` folder and listed first in the font list, with a preview. Each letter is a single stroke: the text is engraved in one pass with a **Profile On the line** toolpath, with a V-bit or a small end mill. It is the fastest way to mark part numbers, plates and assembly marks. When the selection holds only open strokes, **Machine…** offers this profile directly. The `fonts/apercu-polices.html` catalogue (link “See the single-line font catalogue” in the Text block) shows them all with a preview of your text. You can also add your own SVG fonts with **Add fonts…**.

### Decorative fonts

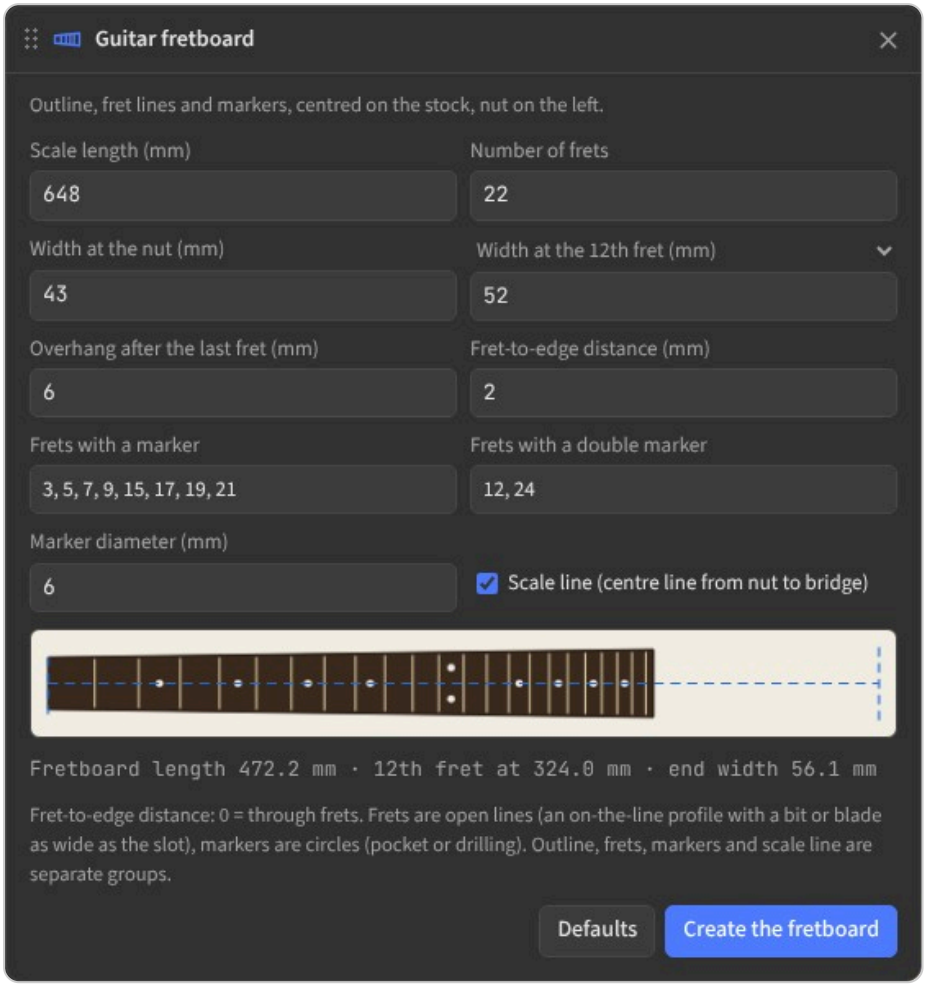
Nine **decorative font** families are also bundled, in the `fonts/decoratives` folder, and listed under *Decorative fonts (bundled with COPO)*: **URW Chancery** (chancery italic), **Alegreya** (roman and italic), **Cinzel Decorative** (ornate capitals, 3 weights), **Pirata One** (blackletter), **Chicle**, **Comic Neue** (6 styles), **Protest Riot**, **Great Vibes** (script) and **Smokum** (western). These are outline fonts: machine them with V-carving, a pocket or a profile. For thin scripts such as Great Vibes, use a large enough text height (20 mm or more) so the bit fits in the hairlines. The *COPO fonts* box hides them together with the single-line fonts. Licences: SIL Open Font License for most, Apache 2.0 for Smokum, AGPL 3 with font exception for URW Chancery (see `fonts/decoratives/LICENCES.txt`).

## Lutherie: fretboard and inlay

### Guitar fretboard

**Gadgets** › **Guitar fretboard** (or Drawing › Guitar fretboard…) draws the fretboard, nut on the left, centred on the stock. Set the **scale length**, the **number of frets**,

the **width at the nut** and the second width, *at the 12th fret* or *at the end of the scale*, the overhang after the last fret, the **frets with a marker** and **a double marker** (lists of numbers) and the marker diameter. The **scale line** (ticked by default) draws the centre line from the nut to the bridge, with a mark at each end, to place the bridge. With the **Zero fret** box, a fret 0 is drawn at the start of the scale (with the same edge distance as the others) and the board starts before it, at the distance set in *Board start → zero fret*; the "nut" width is then the zero-fret width. The **fret-to-edge distance** stops the fret lines short of the board edge (0 = through frets). The preview updates live. The window opens next to the left panel, like the toolpath settings, and moves by its title.




The **Guitar fretboard** window with its preview: board, frets, markers and the dashed scale line.

### Fret positions

COPO uses the exact equal-temperament formula, the one used by reference fret calculators:

> $\mathbf{d_n = L - L / 2^{n/12}}$ i.e. $d_n = L \times (1 - 2^{-n/12})$

where $L$ is the scale length, $n$ the fret number and $d_n$ the distance from the nut to the fret. Each fret shortens the string by a semitone, a factor of $2^{1/12} \approx 1.05946$; the $12^{th}$ fret falls exactly at half the scale. The “rule of 17.817” gives the same result; the old, less accurate rule of 18 is not used. No compensation is applied to the frets: intonation is set at the bridge, moved back a few millimetres from the scale line.

| FRET | 1 | 5 | 7 | 12 | |
|---|---|---|---|---|---|
| Distance from the nut (648 mm scale) | 36.37 mm | 163.51 mm | 215.64 mm | 324.00 mm | 466.1 |

The result is separate groups: the scale line (not to be machined), the outline, the fret lines (open lines, to cut with an **On the line** profile and a bit or blade as wide as the slot) and the markers (circles, pocketed or drilled).

### V-carve inlay

Select the design (closed shapes), then **Drawing › V-carve inlay…**. Choose the V-bit, the **female pocket depth**, the **glue gap** and the **male cut depth**. COPO creates:

- a *female* V-carving of the design in your part;
- the *male* plug: the mirrored design inside a frame, on the “Inlay · male” layer, with a V-carving that starts at the pocket depth minus the gap. Its toolpath is created *hidden*: to cut it from the inlay wood, show it and hide the female one before saving the G-code.

Flip the plug, glue it into the female pocket, then plane or sand off the excess. The V-carving **Start depth** setting is also available by hand in the toolpath form.

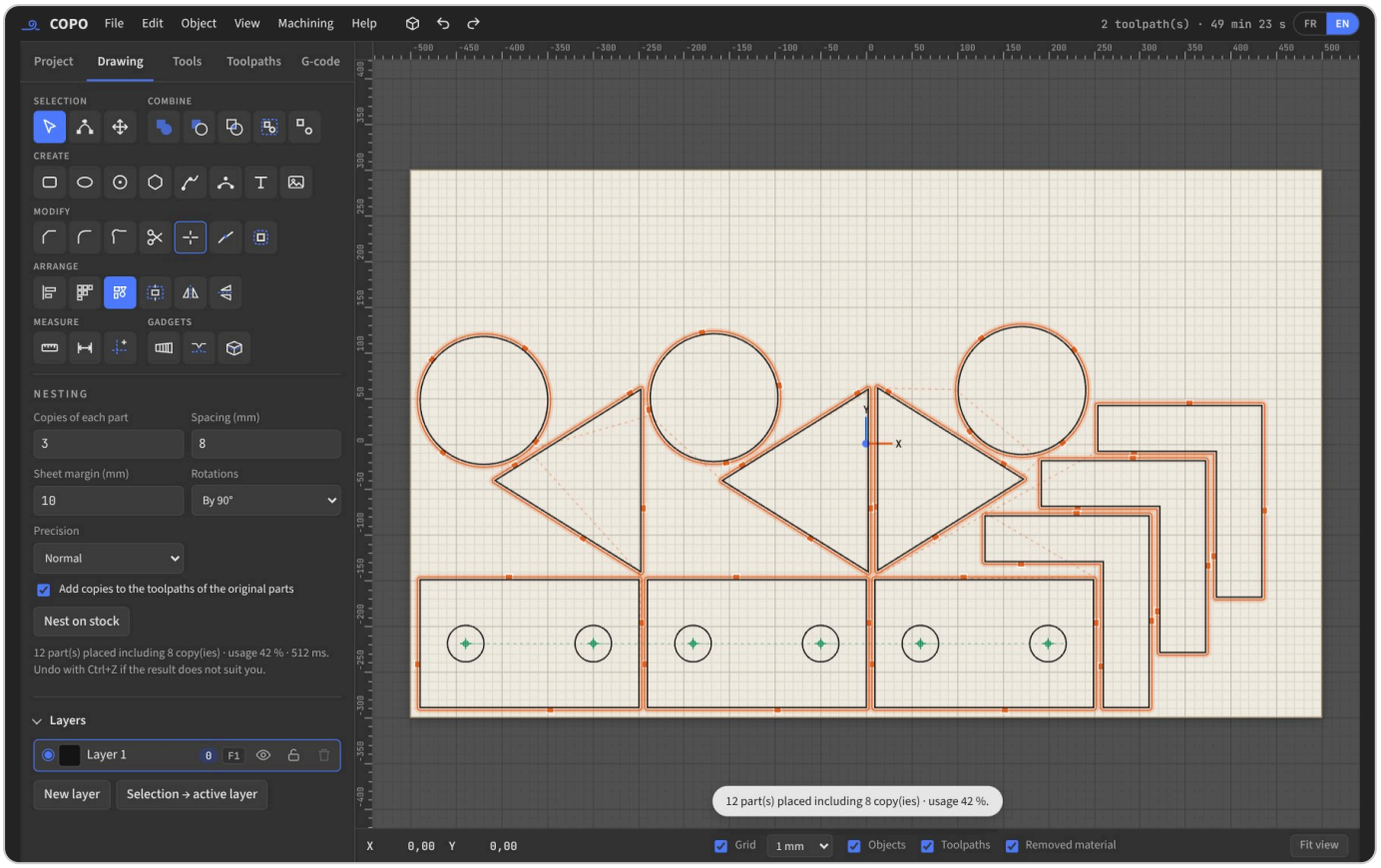
## Finger-joint box

In the **Gadgets** group of the Drawing tab (or **Drawing › Finger-joint box…**), this module draws a box assembled with straight finger joints, like Vectric's “Box Creator”. Set the **outer dimensions** (width X, depth Y, height Z), the **material thickness** (the stock's by default), the **finger width** (their number, always odd, follows), the **joint allowance** (0.1 to 0.2 mm in wood), the **cutting bit**, the **top joint** (*finger-jointed* for a closed box, the default, or *laid flat* for a removable lid) and the **faces to machine** (Top, Bottom, Sides, Ends). Fingers are spread evenly between the corners: the corner zone, as thick as the material, only exists where another panel joins, so an open box keeps the same pitch up to the top. Fingers may be narrower than the thickness; COPO warns when the slots become

narrower than the bit. The *Dogbone reliefs* option adds an arc the size of the bit at the bottom of each slot so square fingers fit.

The preview shows the panels laid flat. **Create the box** places them on the stock, spaced for the bit: cut them with an outside profile and tabs. The panel volumes fit exactly: no overlap or gap at the corners.

# Nesting




Four parts in three copies on a 1000 × 600 mm sheet. The L shapes interlock and the copies join the toolpaths of the original parts.

**Drawing › Nesting…** opens the block in the Drawing tab. Select the parts (with no selection, the whole drawing is used) and set the number of copies, the spacing, the sheet margin, the allowed rotations and the precision. COPO places the parts following their true shape, largest first, as far down and left as possible. A group or a text stays in one block. Parts that do not fit stay selected, and the sheet usage is shown.

> For cut-outs, use a spacing at least equal to the bit diameter. `Ctrl` + `Z` undoes the nesting if you don't like the result.

# Double-sided machining

1. Put the vectors of the bottom side on a layer set to **F2** (F1/F2 button in the layer list). Draw them as seen from above, see-through.
2. Choose the flip: around the X axis (top ↔ bottom) or the Y axis (left ↔ right).
3. Side 2 toolpaths are calculated on the flipped geometry; the cut direction is kept.
4. In the G-code, side 1 is followed by an `M0` pause: turn the part over, set X0 Y0 Z0 again at the same point and resume.

The 3D preview shows both sides; the **Side 2 up** option shows the part flipped.

# 2D and 3D views

The **2D** and **3D** buttons at the top work as two switches: 2D only (the drawing fills the area), 3D only (the relief), or both at once. When both are shown, the buttons next to them place the views side by side or one above the other. The Relief tab shows the 3D view automatically.

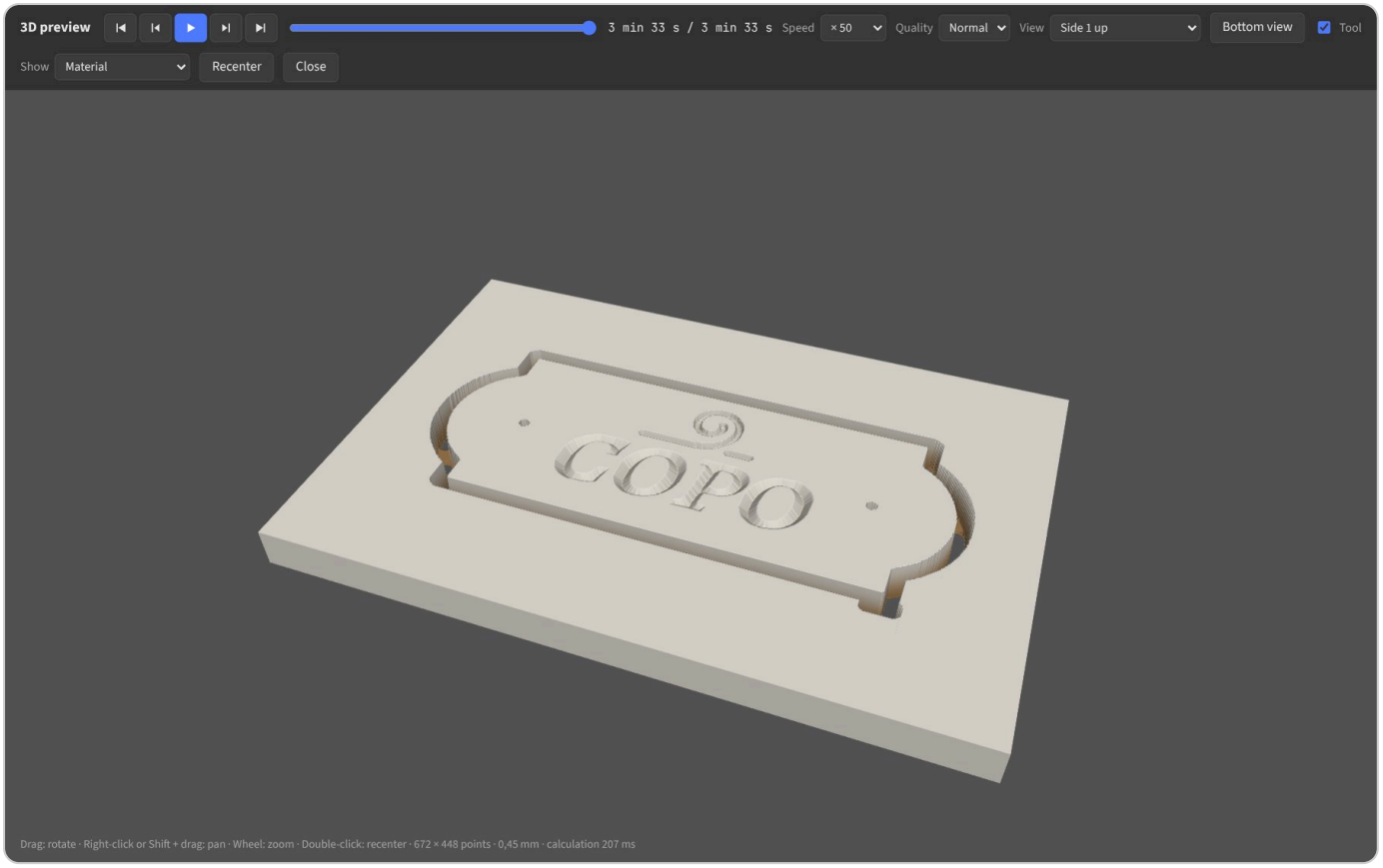
The 3D view shows the relief in the stock (wireframe), with the drawing and the axes. The view uses an orthographic projection (no perspective), handy from the front or the side. Drag: rotate; right-click or Shift + drag: pan; wheel: zoom; double-click: whole stock. Top right: the view cube. At the bottom, icon bars like in the 2D view: on the left, show or hide the drawing, the axes and the stock; in the middle, the transform tools; on the right, recentre on the relief.

**Transforming a relief in 3D.** Click an item in the 3D view to select it (its box corners are highlighted; the selection is shared with the 2D view and the list). The bottom bar offers three tools, with dimensions shown at the top left:

- **Move**: drag the X, Y or Z arrows (Shift: whole millimetres). In Z, the relief top snaps to the stock top and its bottom to the stock bottom; in X and Y, its edges snap to the stock edges.
- **Rotate**: drag the blue ring (around Z); for an STL, the red and green rings also tilt it around X and Y (Shift: 15° steps).
- **Resize**: drag a corner to scale keeping proportions (Shift: freely), a side to change only the width or the length, or the blue dot above for the height: the top moves, the bottom stays put (Shift: whole millimetres).
- The last two buttons **drop the relief onto the stock bottom** or **align its top with the stock top**.

Each transformation can be undone with **Ctrl+Z**; the relief panel keeps the exact values.

# 3D preview




Drag to rotate, right-click or Shift to pan, wheel to zoom, double-click to recenter.

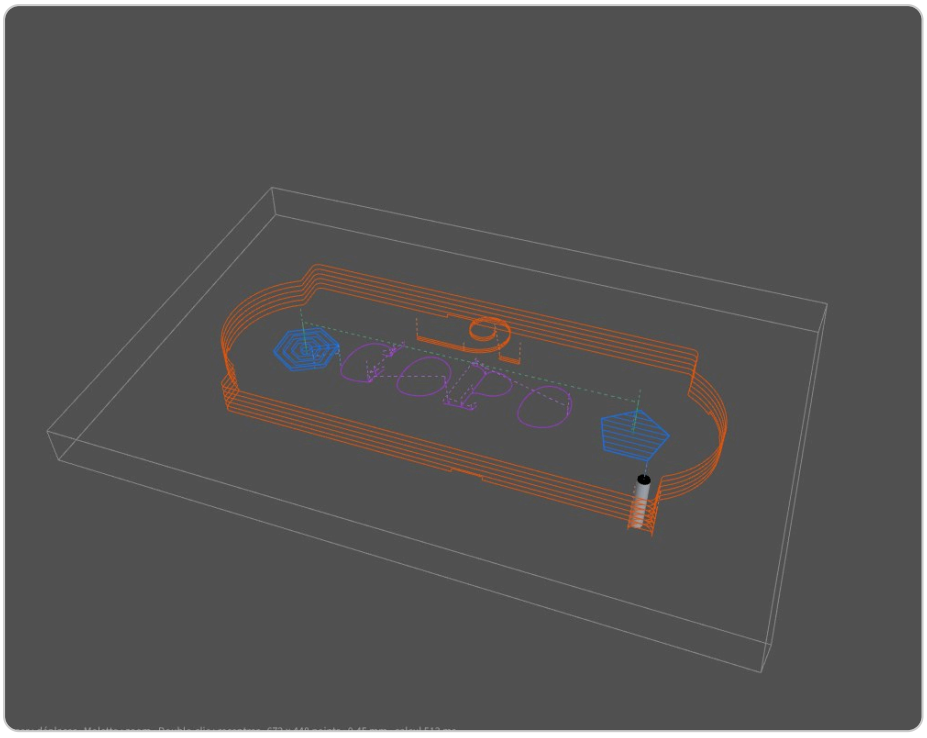
The preview simulates material removal through the real stock thickness, through-holes included. The playback buttons replay the machining: **back to start**, **step back**, **play / pause**, **step forward** and **go to end**; one step is one tool move. Keyboard: left and right arrows, Home, End and Space. The slider jumps to any moment.

**View cube** (top-right corner): it turns with the view and shows the faces (Top, Front, Right…) and the X, Y, Z axes. Click a face for that view (Top = plan view, like the drawing), or a corner for an isometric view. The tool shown keeps its shape: flat, ball-nose, V-bit, drill or thread mill (60° tooth at the tip, relieved neck). Threads appear in the preview as the last pass progresses (ISO 60° profile), inside the hole or around the screw.

**Relief and machined part**: the **Relief** box overlays the designed relief (a line mesh at its real height) to compare it with the machined material. The **Machined part only** box hides the stock the cutter has not touched: only the machined surfaces remain, handy to judge a relief or a pocket without the rest of the board.

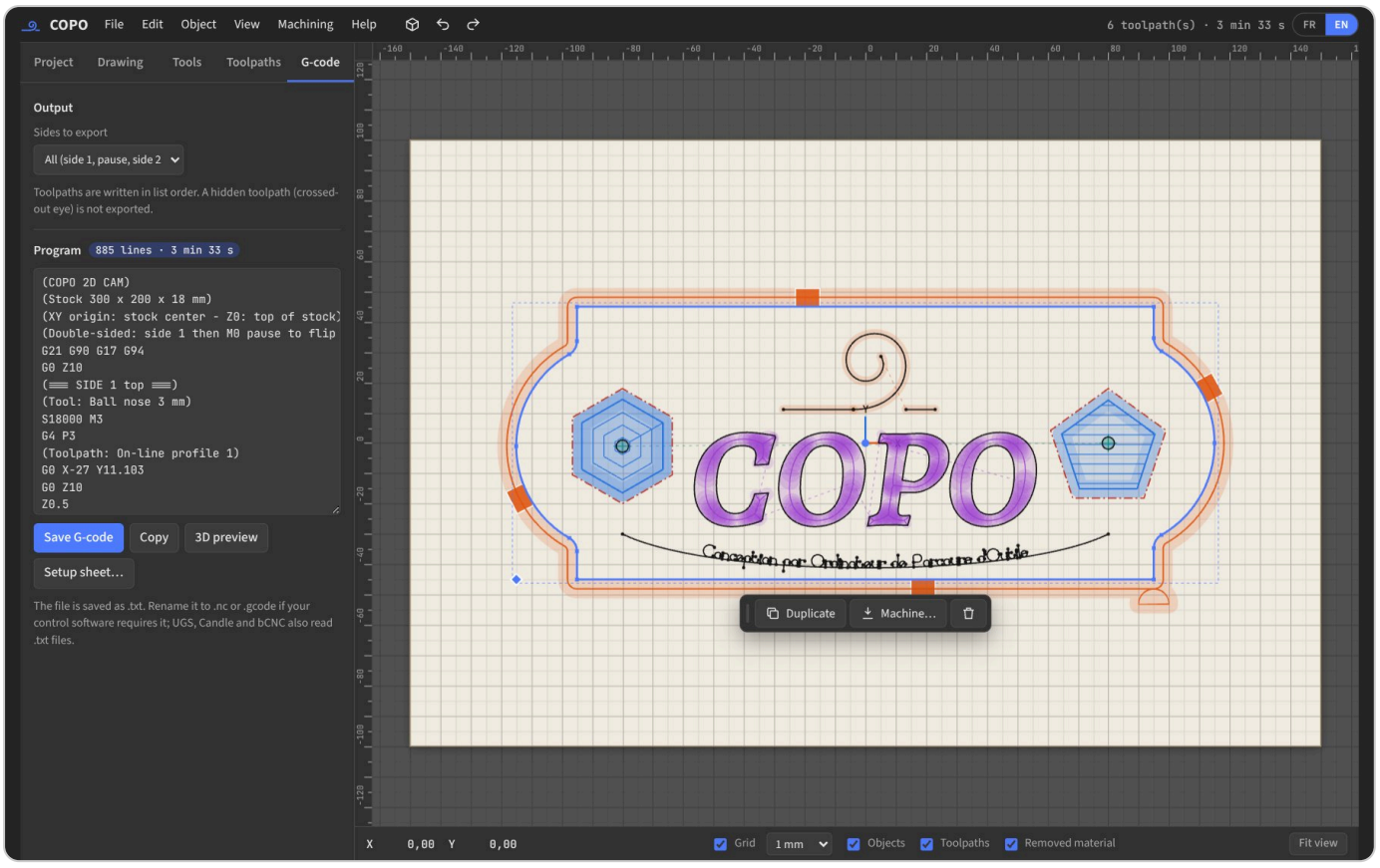
## Toolpaths in 3D

The **Show** list in the preview bar offers *Material*, *Material + toolpaths* or *Toolpaths only*. The **Drawing** box shows or hides the drawing's vectors, in their layer colour, on the face turned upwards: face 1 vectors in the Face 1 view, face 2 vectors in the Face 2 view. Toolpaths are drawn as lines in space, in the colour of their type; rapid moves (G0) are dashed grey. During playback they are drawn as the tool moves. With a toolpath selected in the list, the **Selected toolpath only** box shows just that one. Great for checking passes, ramps, helixes and clearance heights before cutting.



*Toolpaths only* mode: every cut-out pass is visible, rapids are dashed.

# G-code




Toolpaths are written in list order; a hidden toolpath is not exported.

| POST-PROCESSOR | TOOL CHANGE |
|---|---|
| GRBL | `M0` pause between two tools. |
| GRBL manual | Before each tool, the first one included: retract with `G53 G0 Z-10` (adjustable) and `M0` pause. Also a `G53` retract at the end of the program. |
| Carbide Motion, Onefinity | `M6 T…` : the control software asks you to change the tool. |
| LinuxCNC, Mach3/4, UCCNC, MASSO | `T… M6` with the tool number, `G40 G49 G80` header, arcs with `G91.1` . `.ngc` file (LinuxCNC), `.tap` (Mach, UCCNC) or `.nc` . |
| Custom | You write the header, tool change and program end, with variables: `{UNITS}` , `{SAFEZ}` , `{TCZ}` , `{T}` , `{S}` , `{D}` , `{TOOL}` . File extension and `G91.1` are up to you. |

```
(Load tool: End mill 6 mm D6 - set Z zero again then resume)
G21 G90 G17 G94
G53 G0 Z-10
M0
S18000 M3
G4 P3
G0 X10 Y10
G0 Z10
…
M5
G53 G0 Z-10
G0 X0 Y0
M30
```

The post-processor, rapid speed (for the time estimate), safe Z, clearance Z, comments and G2/G3 arcs are set in the Project tab, Machine block. The file gets the extension the post-processor expects ( `.nc` , `.tap` , `.ngc` ).

**Curves and G2/G3 arcs.** COPO does not send Bézier curves to the machine as they are: toolpaths are computed as short segments, then, when “G2/G3 arcs” is ticked, circular portions are recognised and written as true G2/G3 arcs (shorter file, smoother motion). A circle drawn with the Circle tool or a vector smoothed in *Arcs* mode gives exact arcs; a free-form curve is approximated by a series of arcs and lines, within tolerance.

> Always check the G-code with a dry run, tool above the stock, before cutting.

## Units: millimetres or inches

In the Project tab, Units block, choose the **Length units** (millimetres or inches) and the **Feed units** (mm/min, mm/s or in/min). The choice applies everywhere: drawing, rulers, grid (1/16, 1/8, 1/4 and 1 inch), coordinates, dimensions, tools, toolpaths and setup sheet. In inches, one option shows values as fractions (to 1/64) and another writes the G-code in inches ( `G20` ) for controllers set up that way.

The project is always saved in millimetres: you can switch units at any time without distorting anything, and a project opens correctly for someone working in the other unit. The choice is remembered in the browser.

## Setup sheet

The **Setup sheet…** button in the G-code tab (also in the Machining menu) prepares a two-page setup sheet for the workshop. Enter the part name, the

material and your notes; the sheet contains:

- the stock, X0 Y0 origin, Z zero, flip and post-processor;
- a summary: estimated time, maximum depth, machined extent from the origin, cutting length and removed material volume;
- the 2D view of each side with toolpaths, origin and stock dimensions, and two dimetric views of the simulation (side 1 and side 2);
- the tool loading order, then the toolpath table: operation, tool, depth, passes, allowance, entry, tabs, feed, plunge and time;
- the points to watch it detects (cut through the stock, cut-out without tabs, pass too deep, job outside the stock, hidden toolpaths);
- a checklist before starting, a notes area and sign-off boxes.

**Save as PDF** (in the sheet window, or straight from **File** › **Save the setup sheet as PDF** without opening it) creates an A4 PDF file, split into pages without cutting tables or pictures, ready to print or send. The PDF is a picture of the sheet (text not selectable); for a PDF with text, use **Print** then the browser's "Save as PDF". You can also keep the HTML file with the G-code. The sheet settings are saved with the project.

### Send to the control software

The **Send to the control software** block in the G-code tab drops the program straight into a folder: choose the software (gSender, OpenBuilds CONTROL, UGS, CNCjs, Candle, bCNC, Mach3, Mach4, UCCNC, LinuxCNC, MASSO…), then **Choose…** the folder it opens its programs from (once, COPO remembers it). **Send** writes the file with the expected extension (.gcode, .nc, .tap or .ngc). A browser cannot start another program: you still open (or reload) the file in the control software. For a MASSO, choose its USB stick as the folder. Writing to a folder works in Chrome, Edge, Brave and Vivaldi; in Firefox and Safari the program is downloaded with the right extension.

## Install and use offline

- **Online**: open the GitHub Pages site of the COPO repository. Nothing to install.
- **As an app** (Chrome, Edge, Brave, Vivaldi… on Windows, macOS and Linux): **Help** › **Install COPO as an app…** or the install icon in the address bar. COPO then opens in its own window with its icon in the app menu, works **offline** once loaded, and **double-clicking a .copo file** opens it directly (the file icon depends on the system). Updates arrive automatically when online. Firefox and Safari do not offer this install: COPO works normally there in a tab.

- **Offline**: download the complete COPO folder (“Code › Download ZIP” on GitHub) and unzip it. Open `index.html` with a double-click: everything is already in the `lib` folder (calculation libraries and interface fonts) and in the `fonts` folder. Nothing else to install or download.
- **Interface fonts**: the `lib/fonts` folder holds the COPO fonts (Source Sans 3, JetBrains Mono, Barlow Condensed). Offline, the interface, the setup sheet and this guide keep the same look.
- **Engraving fonts**: put your `.ttf`, `.otf` or `.woff` files in a `fonts` folder next to COPO.html and open it once from the File menu.

COPO runs on Windows, macOS and Linux, in a recent Chrome, Edge or Firefox. The list of installed fonts is only available in Chrome and Edge.

# Keyboard shortcuts

| KEY | ACTION | KEY | ACTION |
|---|---|---|---|
| `V` | Select | `Ctrl` + `Z` / `Y` | Undo / Redo |
| `N` | Nodes | `Ctrl` + `S` / `O` | Save / Open a project |
| `H` or `Space` | Pan the view | `Ctrl` + `D` | Duplicate |
| `R` `E` `C` | Rectangle, ellipse, circle | `Ctrl` + `A` | Select all |
| `O` | Polygon and star | `Ctrl` + `G` | Group ( `Shift` : ungroup) |
| `P` `A` `T` | Pen, arc, text | `Esc` / right-click | Back to selection, deselect |
| `K` `F` | Chamfer, fillet | `Del` | Delete the selection or node |
| `X` `U` | Trim, cut | Arrows | Move by 1 mm ( `Shift` : 10 mm) |
| `M` `D` | Measure, dimension | `Esc` | Cancel the current path, deselect |

On a Mac, `Cmd` replaces `Ctrl`.

# Report a bug

**Help › Report a bug…** menu: describe the problem, then click **Open the report on GitHub**. The report arrives pre-filled with technical information (browser, units, stock, toolpaths, recent errors). The workspace screenshot is copied to the clipboard: paste it into the report with `Ctrl` + `V`. If you tick “Attach the

project”, drag the downloaded file into the report. A free GitHub account is enough. Ideas for improvements are welcome too at github.com/gbulleryahen-hash/copo/issues.

**Help › Suggest an improvement…** opens the same window for an idea: describe the feature you want and what it would be used for; it reaches GitHub with the *enhancement* label. Both carry the COPO **version number** (also shown in **Help › About**): 1.*n*, where *n* goes up with every published update.

# Licence

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Libraries used, bundled in the `lib` folder: Clipper (Boost Software License), opentype.js (MIT), d3-delaunay and delaunator (ISC).